

# SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

## Functions

Week 1 Day 2 AM



# OUTLINE

## ① Functions

## ② Graphs of Functions

## ③ Getting Information from the Graph of a Function

## ④ Summary

# MOTIVATIONS

In scientific and engineering problems, we often want to study quantities that depend on other quantities. For instance,

- How does the position of the moon vary with time?
- How does the temperature change with altitude?
- How does the power output of an engine depend on the oxygen-fuel ratio?
- How does the computational time of a computer program depend on the size of the input data?

The idea of the dependency of one quantity upon another quantity is captured by functions, which are the fundamental objects that we'll study in calculus. After all, calculus is the study of changes: how does the change in one quantity affect the change in another quantity. So let's look at functions formally.

# DOMAIN, CODOMAIN AND RANGE

## DEFINITION:

A **function**  $f : A \longrightarrow B$  is a rule to assign each element  $x$  in set  $A$  to **exactly one** corresponding value in set  $B$ , denoted as  $f(x)$ .  $A$  is called the **domain** of  $f$ ,  $B$  is called the **codomain** of  $f$ ,  $x$  is called the **independent variable**/**input**/**preimage of  $f(x)$** , and  $f(x)$  is called the **dependent variable**/**output**/**image of  $x$  under  $f$** .

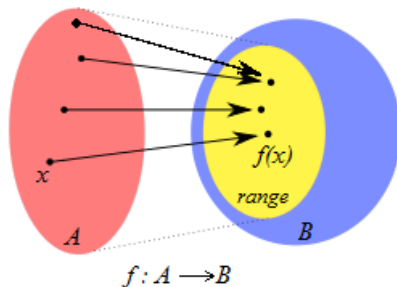
⚠  $A, B$  can be any sets, but we'll restrict them to subsets of  $\mathbb{R}$  for this course.

## DEFINITION:

The subset of  $B$  that contains all the  $f(x)$  is called the **range** of  $f$ , i.e.

$$\text{range of } f := \{y \in B \mid \text{there exists } x \in A \text{ such that } y = f(x)\}.$$

# EXAMPLES OF FUNCTIONS



**E.g.:**

$$f: \mathbb{R} \longrightarrow \mathbb{R}$$

$$x \longmapsto x^2$$

The domain of  $f$  is  $\mathbb{R}$ , the codomain of  $f$  is  $\mathbb{R}$  and the rule of  $f$  is to square the input. The range of  $f$  is  $[0, \infty)$ .

# EXAMPLES OF FUNCTIONS

**E.g.:**

$$g : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$$

$$x \longmapsto \frac{1}{x}$$

The domain of  $g$  is  $\mathbb{R} \setminus \{0\}$ , the codomain of  $g$  is  $\mathbb{R}$  and the range of  $g$  is  $\mathbb{R} \setminus \{0\}$ .

**E.g.:**  $h(x) = \begin{cases} 1 & \text{if } x \in \mathbb{R} \text{ is a rational number,} \\ 0 & \text{if } x \in \mathbb{R} \text{ is irrational.} \end{cases}$

The domain of  $h$  is  $\mathbb{R}$ , the codomain is not stated and the range of  $h$  is  $\{0, 1\}$ . This is an example of a piecewise defined function, i.e. the rule is split into pieces according to subsets of the domain.

**E.g.:**  $j(x) = \pm\sqrt{x}$ , for  $x \in [0, \infty)$ .  $j$  is **not** a function because it assigns **two values** for each  $x > 0$ .

# EVALUATING A FUNCTION

Given a function  $f(x)$ , we evaluate  $f(\square)$  by substituting  $x = \square$  in the rule of  $f$ .

**E.g.:**

$$\begin{aligned} f : \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\longmapsto x^2 - 2x \end{aligned}$$

The rule of  $f$  is  $f(x) = x^2 - 2x$ . Then

- $f(-1) = (-1)^2 - 2(-1) = 3$ .
- $f(y) = y^2 - 2y$ .
- $f(x^2) = (x^2)^2 - 2x^2 = x^4 - 2x^2$ .
- $f(1+t) = (1+t)^2 - 2(1+t) = t^2 + 2t + 1 - 2 - 2t = t^2 - 1$ .
- $f(a+b) = (a+b)^2 - 2(a+b)$ .

# REMARKS

A function consists of 3 pieces of information: domain, codomain and the rule. Two functions  $f, g$  are equal iff. they have the same domain, codomain and rule. For instance, let

$$f : \mathbb{R} \longrightarrow \mathbb{R}$$

$$x \longmapsto x^2$$

$$g : [0, \infty) \longrightarrow \mathbb{R}$$

$$x \longmapsto x^2.$$

Then  $f \neq g$ .

Although we normally denote the independent variable by  $x$ , we can use other symbols such as  $y$  and  $t$ . For instance, the following two functions are equal:

$$f : \mathbb{R} \longrightarrow \mathbb{R}$$

$$x \longmapsto x^2$$

$$h : \mathbb{R} \longrightarrow \mathbb{R}$$

$$y \longmapsto y^2.$$

# CONVENTION ON THE DOMAIN AND CODOMAIN

If only the rule of the function  $f(x)$  is given without any context, domain and codomain, the convention (for this course) is that the domain of  $f$  is **the largest subset of  $\mathbb{R}$  that contains all the  $x$  such that the rule is well-defined**, and the codomain of  $f$  is  $\mathbb{R}$ .

**E.g.:**  $f(x) = \frac{1}{x-1}$ . By convention, the domain of  $f$  is  $\mathbb{R} \setminus \{1\}$  and the codomain is  $\mathbb{R}$ .

**E.g.:**  $g(x) = \sqrt{x}$ . By convention, the domain of  $g$  is  $[0, \infty)$  and the codomain is  $\mathbb{R}$ .

# ACTIVITY 1

What is the domain and codomain of the following functions?  
Evaluate the functions at  $-2, -1, 0, 1, 2$  whenever these numbers are in their domains.

(a)

$$f : [0, 10] \longrightarrow [0, \infty)$$

$$x \longmapsto 2x + 1$$

(b)  $g(x) = \sqrt{x - 1}$

(c)  $h(t) = \frac{t+1}{t-1}$

(d)  $k(x) = \frac{|x|}{x}$

(e)  $p(x) = \begin{cases} x^2 + 3x & \text{if } x < -1, \\ x & \text{if } x \geq -1. \end{cases}$

# ACTIVITY 1: SOLUTION

- (a) Domain =  $[0, 10]$ , codomain =  $[0, \infty)$ .  $-2, -1 \notin$  domain of  $f$ .

$$f(0) = 2(0) + 1 = 1,$$

$$f(1) = 2(1) + 1 = 3,$$

$$f(2) = 2(2) + 1 = 5.$$

- (b) Since  $\sqrt{x-1}$  is defined iff.  $x-1 \geq 0$ , the domain is  $[1, \infty)$  and the codomain is  $\mathbb{R}$  by convention.  $-2, -1, 0 \notin$  domain of  $g$ .

$$g(1) = \sqrt{1-1} = 0,$$

$$g(2) = \sqrt{2-1} = \sqrt{1} = 1.$$

- (c) Since  $\frac{t+1}{t-1}$  is defined iff.  $t-1 \neq 0$ , the domain is  $\mathbb{R} \setminus \{1\}$  and the codomain is  $\mathbb{R}$  by convention.  $1 \notin$  domain of  $h$ .

$$h(-2) = \frac{-2+1}{-2-1} = \frac{1}{3},$$

$$h(-1) = \frac{-1+1}{-1-1} = 0,$$

$$h(0) = \frac{0+1}{0-1} = -1,$$

$$h(2) = \frac{2+1}{2-1} = 3.$$

# ACTIVITY 1: SOLUTION

- (d) Since  $\frac{|x|}{x}$  is defined iff.  $x \neq 0$ , the domain is  $\mathbb{R} \setminus \{0\}$  and the codomain is  $\mathbb{R}$  by convention.  $0 \notin \text{domain of } k$ .

$$k(-2) = \frac{|-2|}{-2} = \frac{2}{-2} = -1,$$

$$k(-1) = \frac{|-1|}{-1} = -1,$$

$$k(1) = \frac{|1|}{1} = 1,$$

$$k(2) = \frac{|2|}{2} = 1.$$

- (e) Since the pieces are defined for  $x < -1$  and  $x \geq -1$ , their union  $= \mathbb{R}$  is the domain. The codomain is  $\mathbb{R}$  by convention.

Since  $-2 < -1$ , the rule is  $p(x) = x^2 + 3x$ , so

$$p(-2) = (-2)^2 + 3(-2) = -2.$$

On the other hand, since  $-1, 0, 1, 2 \geq -1$ , the rule is

$p(x) = x$ , so

$$p(-1) = -1, p(0) = 0,$$

$$p(1) = 1, p(2) = 2.$$

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# GRAPHS OF FUNCTIONS

## DEFINITION:

If  $f$  is a function with domain  $A$ , then the **graph** of  $f$  is the set of ordered pairs

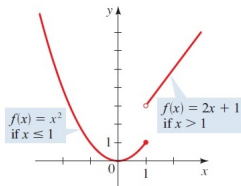
$$\{(x, f(x)) \mid x \in A\}$$

plotted in a coordinate plane. In other words, the graph of  $f$  is the graph of the equation  $y = f(x)$ .

**Notations:** An empty circle on a graph is used to indicate that the point is **not** part of the graph. A filled circle on a graph is used to indicate that the point is part of the graph.

**E.g.:**

$$f(x) = \begin{cases} x^2 & \text{if } x \leq 1, \\ 2x + 1 & \text{if } x > 1. \end{cases}$$

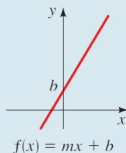
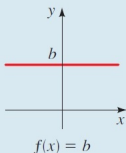


# GRAPHS OF FUNCTIONS - EXAMPLES

## SOME FUNCTIONS AND THEIR GRAPHS

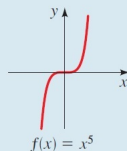
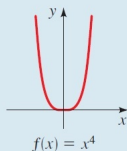
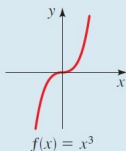
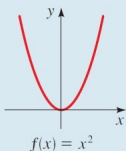
### Linear functions

$$f(x) = mx + b$$



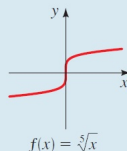
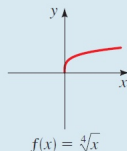
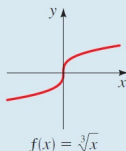
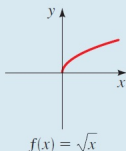
### Power functions

$$f(x) = x^n$$



### Root functions

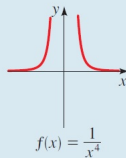
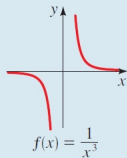
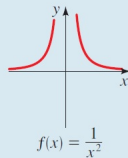
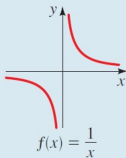
$$f(x) = \sqrt[n]{x}$$



# GRAPHS OF FUNCTIONS - EXAMPLES

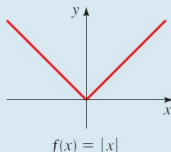
## Reciprocal functions

$$f(x) = \frac{1}{x^n}$$



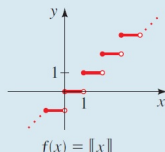
## Absolute value function

$$f(x) = |x|$$



## Greatest integer function

$$f(x) = \llbracket x \rrbracket$$



**E.g.:** The greatest integer function is defined as

$\llbracket x \rrbracket$  or  $\llbracket x \rrbracket :=$  greatest integer less than or equal to  $x$ , for  $x \in \mathbb{R}$ .

For example,  $\llbracket 2.3 \rrbracket = 2$ ,  $\llbracket \pi \rrbracket = 3$ ,  $\llbracket 7 \rrbracket = 7$ ,  $\llbracket -0.1 \rrbracket = -1$  (it's not 0, as  $0 > -0.1$ ).

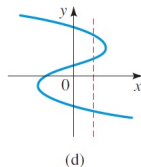
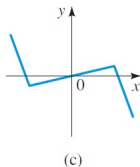
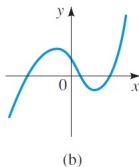
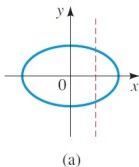
# VERTICAL LINE TEST

Since for any  $x$  in the domain of a function  $f$ , there's exactly **one** corresponding  $f(x)$  value, this translates to the following geometrical property of the graph of a function.

## THE VERTICAL LINE TEST:

A curve in the coordinate plane is the graph of a function if and only if no vertical line intersects the curve more than once.

**E.g.:** By the vertical line test, curves (a) and (d) are not the graphs of some functions, whereas curves (b) and (c) are the graphs of some functions.



## ACTIVITY 2

Sketch the graphs of the following functions.

(a)  $f(x) = 2x + 1, \quad 0 \leq x \leq 10.$

(b)  $k(x) = \frac{|x|}{x}, \quad x \in \mathbb{R} \setminus \{0\}.$

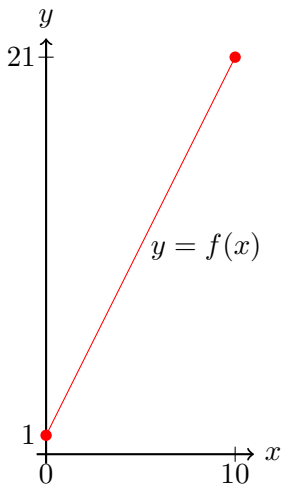
(c)  $p(x) = \begin{cases} x^2 + 3x & \text{if } x < -1, \\ x & \text{if } x \geq -1. \end{cases}$

Hint: complete the square.

Note that it's impossible to sketch graphs over an infinite domain. In this case, we simply sketch parts of the graph that show all the important features.

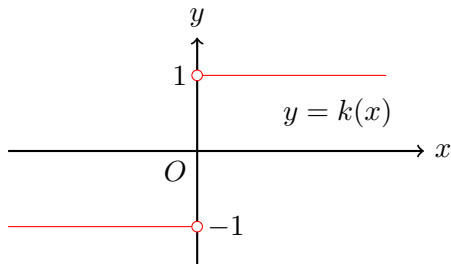
## ACTIVITY 2: SOLUTION

- (a) This is the line with slope 2 and  $y$ -intercept 1, restricted to the domain  $[0, 10]$ .



# ACTIVITY 2: SOLUTION

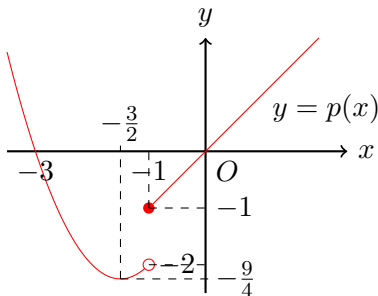
(b) For  $x < 0$ ,  $k(x) = \frac{|x|}{x} = \frac{-x}{x} = -1$ . For  $x > 0$ ,  
 $k(x) = \frac{|x|}{x} = \frac{x}{x} = 1$ .



## ACTIVITY 2: SOLUTION

(c) For  $x < -1$ ,  $p(x) = x^2 + 3x = x^2 + 3x + \frac{9}{4} - \frac{9}{4}$   
 $= \left(x + \frac{3}{2}\right)^2 - \frac{9}{4}$ . Thus this part of the graph is a  $\cup$  shape  
 with the lowest point at  $\left(-\frac{3}{2}, -\frac{9}{4}\right)$ .

For  $x \geq -1$ ,  $p(x) = x$ , the line with slope 1 passing through the origin.



It's hard to sketch graphs by hand in general. It will become easier once you have learned differential calculus.

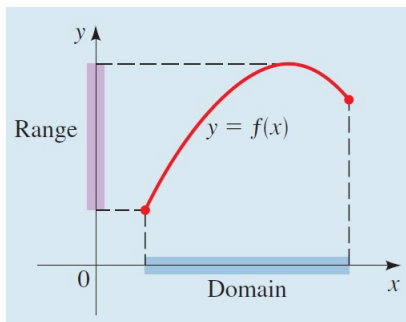
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# FINDING THE RANGE

Many properties of a function are more easily obtained from a graph than from the rule that describes the function.

The range of a function  $y = f(x)$  with a given domain can be obtained from the graph of  $f$  as shown in the figure below. The range is all the corresponding  $y$ -values of the graph over the domain.



# ACTIVITY 3

What is the range of the following functions from activity 1? You may use [Geogebra](#) to sketch some of the graphs.

(a)

$$\begin{aligned} f : [0, 10] &\longrightarrow [0, \infty) \\ x &\longmapsto 2x + 1 \end{aligned}$$

(b)  $g(x) = \sqrt{x - 1}$

(c)  $h(t) = \frac{t+1}{t-1}$

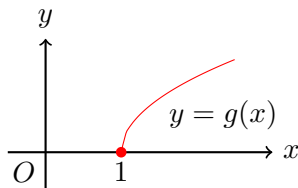
(d)  $k(x) = \frac{|x|}{x}$

(e)  $p(x) = \begin{cases} x^2 + 3x & \text{if } x < -1, \\ x & \text{if } x \geq -1. \end{cases}$

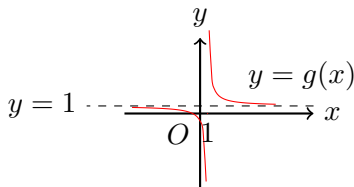
# ACTIVITY 3: SOLUTION

(a) The range of  $f$  is  $[1, 21]$ .

(b) The range of  $g$  is  $[0, \infty)$ .

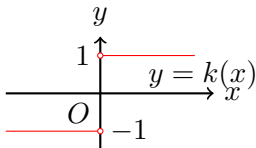


(c) The range of  $h$  is  $\mathbb{R} \setminus \{1\}$ .

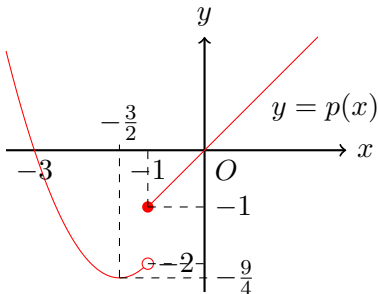


## ACTIVITY 3: SOLUTION

(d) The range of  $k$  is  $\{-1, 1\}$ .



(e) The range of  $p$  is  $\left[-\frac{9}{4}, \infty\right)$ .



# COMPARING FUNCTIONS VALUES FOR SOLVING EQUALITIES OR INEQUALITIES

We can compare the values of two functions  $f$  and  $g$  visually by drawing their graphs.

The points at which the graphs intersect are the points where the values of the two functions are equal. So the solutions of the equation  $f(x) = g(x)$  are the values of  $x$  at which the two graphs intersect.

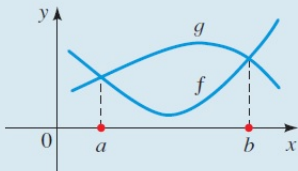
The points at which the graph of  $g$  is higher than the graph of  $f$  are the points where the values of  $g$  are greater than the values of  $f$ . So the solutions of the inequality  $f(x) < g(x)$  are the values of  $x$  at which the graph of  $g$  is higher than the graph of  $f$ .

# COMPARING FUNCTIONS VALUES FOR SOLVING EQUALITIES OR INEQUALITIES

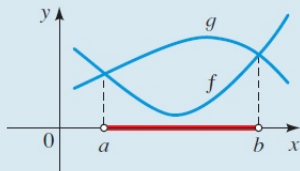
## SOLVING EQUATIONS AND INEQUALITIES GRAPHICALLY

The **solution(s) of the equation**  $f(x) = g(x)$  are the values of  $x$  where the graphs of  $f$  and  $g$  intersect.

The **solution(s) of the inequality**  $f(x) < g(x)$  are the values of  $x$  where the graph of  $g$  is higher than the graph of  $f$ .



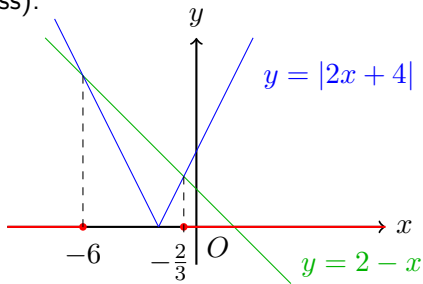
The solutions of  $f(x) = g(x)$   
are the values  $a$  and  $b$ .



The solution of  $f(x) < g(x)$   
is the interval  $(a, b)$ .

# COMPARING FUNCTIONS VALUES FOR SOLVING EQUALITIES OR INEQUALITIES - EXAMPLE

**E.g.:** To solve  $3 - |2x + 4| \leq x + 1 \Leftrightarrow 2 - x \leq |2x + 4|$  from activity 3 of the previous class, we can first sketch the graphs of  $y = 2 - x$  and  $y = |2x + 4|$ . For the graph of  $y = |2x + 4|$ , we simply sketch  $y = 2x + 4$  and then reflect the part below the  $x$ -axis, to make all the  $y$  values positive (we'll talk more on this next class).



Hence, the solution set is  $(-\infty, -6] \cup [-\frac{2}{3}, \infty)$ .

# ACTIVITY 4

Solve the inequalities of activity 3 of the previous class by the graphical method. You may use [Geogebra](#) to sketch some of the graphs.

(a)  $2(7x - 3) \leq 12x + 16$

(b)  $-1 < 2x - 5 < 7$

(c)  $x^2 + 2x > 3$

(d)  $x^2(x^2 - 9) \leq 0$

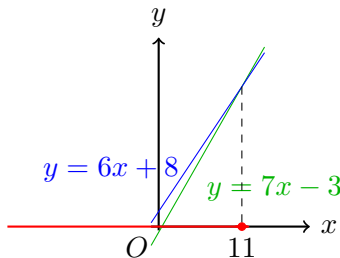
(e)  $\frac{x}{x+1} > 3x$

(f)  $\frac{x+2}{x+3} \leq \frac{x-1}{x-2}$

(g)  $|3x + 5| \leq x - 4$

## ACTIVITY 4: SOLUTION

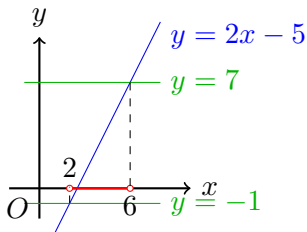
- (a) To solve  $2(7x - 3) \leq 12x + 16 \Leftrightarrow 7x - 3 \leq 6x + 8$ , we look for  $x$  such that the graph of  $y = 7x - 3$  is below or coincides with the graph of  $y = 6x + 8$ .



Hence, the solution set is  $(-\infty, 11]$ .

## ACTIVITY 4: SOLUTION

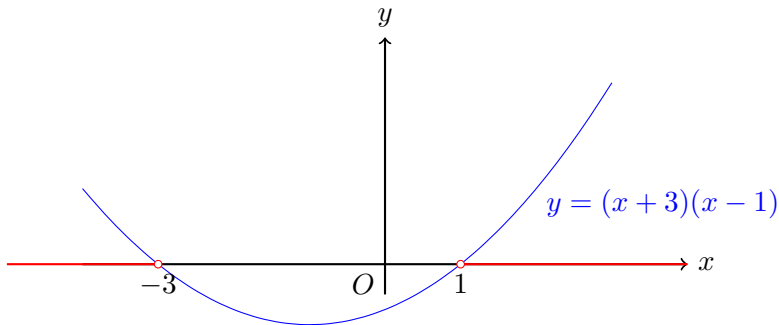
- (b) For  $-1 < 2x - 5 < 7$ , we sketch the graphs  $y = -1$ ,  $y = 7$  and  $y = 2x - 5$ . We need to look for  $x$  such that the  $y$ -value of the graph of  $y = 2x - 5$  is in between the two horizontal lines  $y = -1$  and  $y = 7$ .



Hence, the solution set is  $(2, 6)$ .

## ACTIVITY 4: SOLUTION

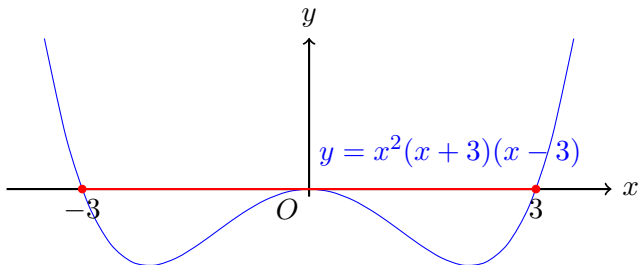
- (c) To solve  $x^2 + 2x > 3 \Leftrightarrow (x + 3)(x - 1) > 0$ , we look for  $x$  such that the graph of  $y = (x + 3)(x - 1)$  is above the line  $y = 0$ , i.e. the  $x$ -axis.



So the solution set is  $(-\infty, -3) \cup (1, \infty)$ .

## ACTIVITY 4: SOLUTION

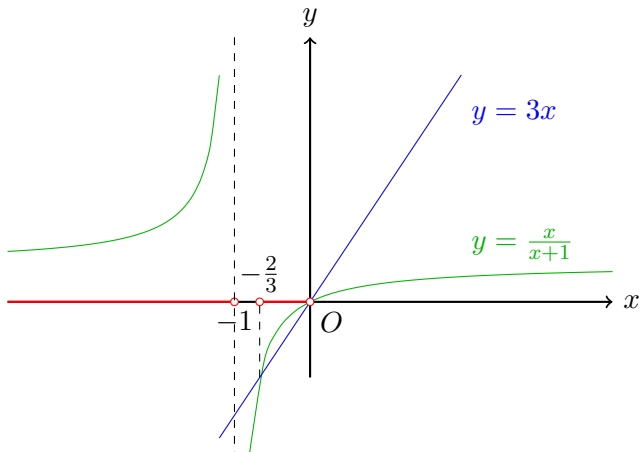
- (d) To solve  $x^2(x^2 - 9) \leq 0 \Leftrightarrow x^2(x + 3)(x - 3) \leq 0$ , we look for  $x$  such that the graph of  $y = x^2(x + 3)(x - 3)$  is below or coincides with the line  $y = 0$ , i.e. the  $x$ -axis.



So the solution set is  $[-3, 3]$ .

# ACTIVITY 4: SOLUTION

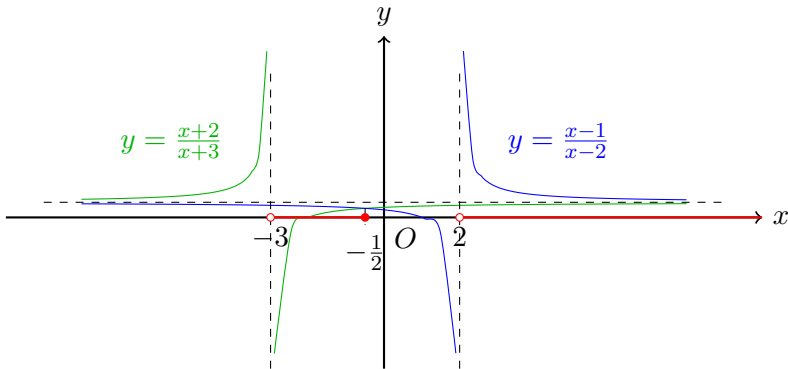
- (e) To solve  $\frac{x}{x+1} > 3x$ , we look for  $x$  such that the graph of  $\frac{x}{x+1}$  is above the graph of  $y = 3x$ .



So the solution set is  $(-\infty, -1) \cup \left(-\frac{2}{3}, 0\right)$ .

## ACTIVITY 4: SOLUTION

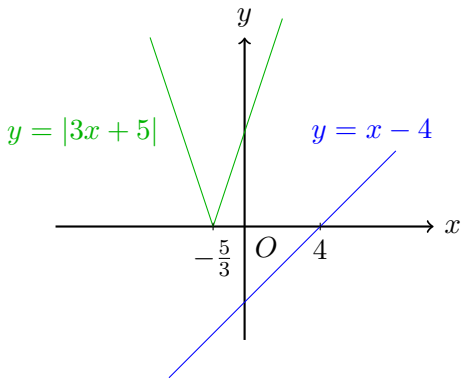
- (f) To solve  $\frac{x+2}{x+3} \leq \frac{x-1}{x-2}$ , we look for  $x$  such that the graph of  $\frac{x+2}{x+3}$  is below or coincides with the graph of  $\frac{x-1}{x-2}$ .



So the solution set is  $\left(-3, -\frac{1}{2}\right] \cup (2, \infty)$ .

## ACTIVITY 4: SOLUTION

- (g) To solve  $|3x + 5| \leq x - 4$ , we look for  $x$  such that the graph of  $y = |3x + 5|$  is below or coincides with the graph of  $y = x - 4$ .

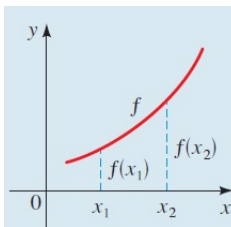


Since the graph of  $y = |3x + 5|$  is always above the graph of  $y = x - 4$ , the solution set is  $\emptyset$ .

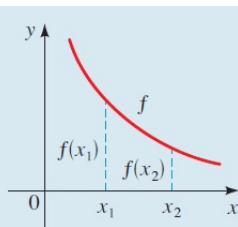
# INCREASING AND DECREASING

## DEFINITION:

- A function  $f$  is **(strictly) increasing** on an interval  $I$  if  $f(x_1) < f(x_2)$  whenever  $x_1 < x_2$  in  $I$ , i.e.  $f(x)$  increases when  $x$  increases.
- A function  $f$  is **(strictly) decreasing** on an interval  $I$  if  $f(x_1) > f(x_2)$  whenever  $x_1 < x_2$  in  $I$ , i.e.  $f(x)$  decreases when  $x$  increases.



$f$  is increasing



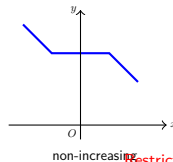
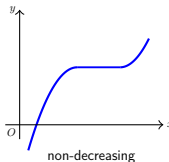
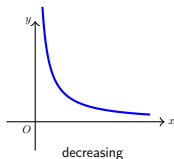
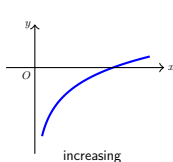
$f$  is decreasing

# NON-INCREASING AND NON-DECREASING

If the inequalities are not strict, then we call them non-increasing/non-decreasing:

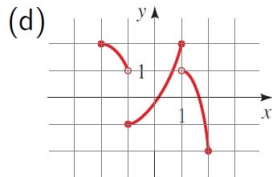
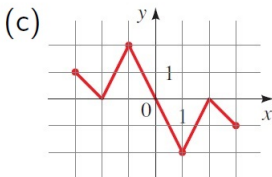
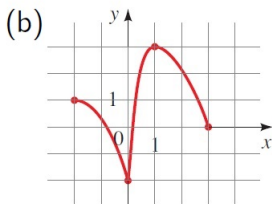
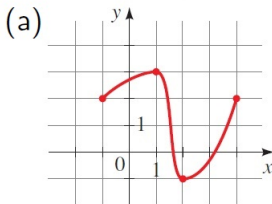
## DEFINITION:

- A function  $f$  is **non-decreasing** on an interval  $I$  if  $f(x_1) \leq f(x_2)$  whenever  $x_1 < x_2$  in  $I$ , i.e.  $f(x)$  stays constant or increases (doesn't decrease) when  $x$  increases.
- A function  $f$  is **non-increasing** on an interval  $I$  if  $f(x_1) \geq f(x_2)$  whenever  $x_1 < x_2$  in  $I$ , i.e.  $f(x)$  stays constant or decreases (doesn't increase) when  $x$  increases.



# ACTIVITY 5 [DO IT AT HOME]

In each of the following part, the complete graph of a function is given. Use the graph to estimate the following. (i) The domain and range of the function. (ii) The intervals on which the function is increasing and on which the function is decreasing.



# ACTIVITY 5: SOLUTION

(a)

- (i) Domain =  $[-1, 4]$ , range =  $[-1, 3]$ .
- (ii) Increasing over  $[-1, 1]$  and  $[2, 4]$ . Decreasing over  $[1, 2]$ .

(b)

- (i) Domain =  $[-2, 3]$ , range =  $[-2, 3]$ .
- (ii) Increasing over  $[0, 1]$ . Decreasing over  $[-2, 0]$  and  $[1, 3]$ .

(c)

- (i) Domain =  $[-3, 3]$ , range =  $[-2, 2]$ .
- (ii) Increasing over  $[-2, -1]$  and  $[1, 2]$ . Decreasing over  $[-3, -2]$ ,  $[-1, 1]$  and  $[2, 3]$ .

(d)

- (i) Domain =  $[-2, 2]$ , range =  $[-2, 2]$ .
- (ii) Increasing over  $[-1, 1]$ . Decreasing over  $[-2, -1]$ , and  $[1, 2]$ .

# GLOBAL EXTREMA

## DEFINITION:

- A function  $f : D \rightarrow \mathbb{R}$  has a **global maximum** at  $a \in D$  if
$$f(a) \geq f(x) \text{ for all } x \in D.$$

Such  $a$  is called a **global maximizer** of  $f$  and  $f(a)$  is called the **global maximum value** of  $f$ .

- A function  $f : D \rightarrow \mathbb{R}$  has a **global minimum** at  $b \in D$  if
$$f(b) \leq f(x) \text{ for all } x \in D.$$

Such  $b$  is called a **global minimizer** of  $f$  and  $f(b)$  is called the **global minimum value** of  $f$ .

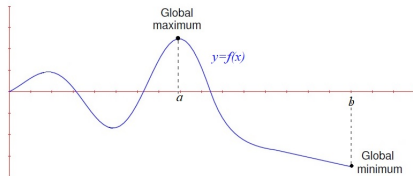


Figure: The graph is drawn over the entire domain.

# GLOBAL EXTREMA

## Remarks:

- Global maximum or global minimum are called *global extremum*.
- Notice the non-strict inequality signs in the definition. Hence, it's possible for the global maximum (or minimum) value to occur at multiple  $x$  values.
- Some functions do not have a global maximum or a global minimum, for example  $f(x) = x^2$  over  $\mathbb{R}$  has no global maximum;  $g(x) = x$  over  $\mathbb{R}$  has neither global maximum nor minimum.
- The branch of mathematics that studies global extrema is called *optimization*. One of the applications of differential calculus is optimization.

# LOCAL EXTREMA

Very often, the strategy to find global extrema is to first find the **local extrema**, which we'll define now:

## DEFINITION:

- A function  $f : D \rightarrow \mathbb{R}$  has a **local maximum** at  $a \in D$  if
$$f(a) \geq f(x) \text{ for all } x \text{ near } a.$$

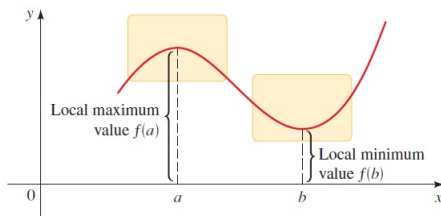
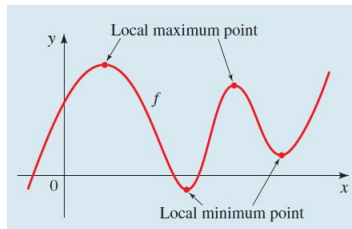
Such  $a$  is called a **local maximizer** of  $f$  and  $f(a)$  is called a **local maximum value** of  $f$ .

- A function  $f : D \rightarrow \mathbb{R}$  has a **local minimum** at  $a \in D$  if
$$f(b) \leq f(x) \text{ for all } x \text{ near } b.$$

Such  $b$  is called a **local minimizer** of  $f$  and  $f(b)$  is called a **local minimum value** of  $f$ .

Remark: By all  $x$  near  $a$ , we mean all  $x$  in the domain  $D$  and in an **open interval  $I$  containing  $a$** , i.e. all  $x \in D \cap I$ .

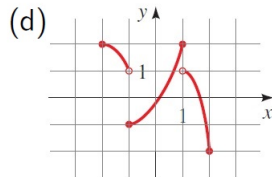
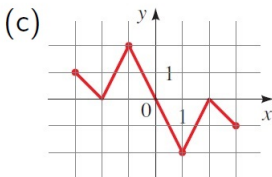
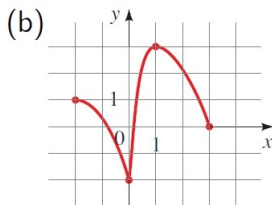
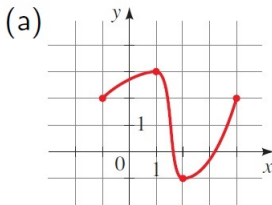
# LOCAL EXTREMA



⚠ The global and local maximizer/minimizer must be an element of the domain of the function!

# ACTIVITY 6

Continuing from the previous activity, find all the global and local extrema of the functions. Note that global extrema are automatically local extrema by definition, but the other way round is not true.



## ACTIVITY 6: SOLUTION

- (a) Global maximum at  $(1, 3)$ .  
Global minimum at  $(2, -1)$ .  
Local maximum at  $(1, 3), (4, 2)$ .  
Local minimum at  $(-1, 2), (2, -1)$ .
- (b) Global maximum at  $(1, 3)$ .  
Global minimum at  $(0, -2)$ .  
Local maximum at  $(-2, 1), (1, 3)$ .  
Local minimum at  $(0, -2), (3, 0)$ .
- (c) Global maximum at  $(-1, 2)$ .  
Global minimum at  $(1, -2)$ .  
Local maximum at  $(-3, 1), (-1, 2), (2, 0)$ .  
Local minimum at  $(-2, 0), (1, -2), (3, -1)$ .
- (d) Global maximum at  $(-2, 2)$  and  $(1, 2)$ .  
Global minimum at  $(2, -2)$ .  
Local maximum at  $(-2, 2), (1, 2)$ .  
Local minimum at  $(-1, -1), (2, -2)$ .

# OUTLINE

- ① Functions
- ② Graphs of Functions
- ③ Getting Information from the Graph of a Function
- ④ Summary

# SUMMARY

- Functions
  - domain, codomain, range.
  - evaluating a function.
  - convention on the domain and codomain.
- Graphs
  - graphs of functions.
  - vertical line test.
- Getting Information from the Graph of a Function
  - finding the range from the graph.
  - comparing functions values for solving equalities or inequalities.
  - increasing, decreasing, non-increasing and non-decreasing.
  - global extrema, local extrema.

# RESOURCES

## References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 2.1, 2.2, 2.3. Sections 2.4, 2.5 are not covered in this course, but it's good to know.